

# Learning Objectives

- Quantify heat transfer from one object to another using  $q = mC_s\Delta T$  or  $q = C\Delta T$ .
- Quantify heat transfer for a chemical reaction.
- Understand the concept of Pressure-Volume work
- Determine the sign of P-V work under conditions of gas expansion/gas contraction.
- Relate the amount of heat released or absorbed by a reaction using a balanced chemical reaction and the reaction stoichiometry.
- Quantify the  $\Delta H$  for a chemical reaction using data from a coffee cup calorimeter.

# The First Law of Thermodynamics

*The law of conservation of energy:*

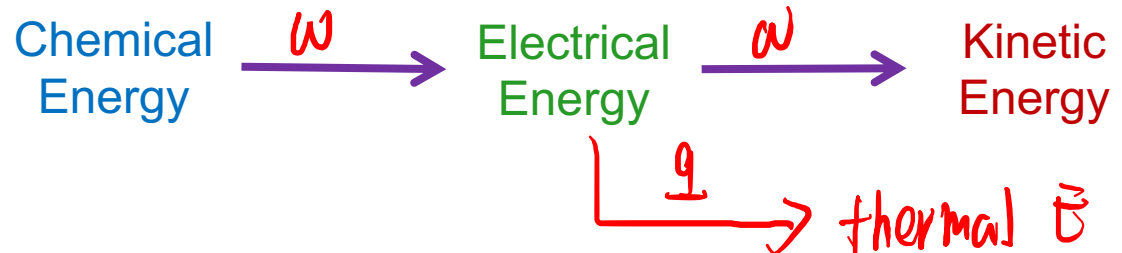
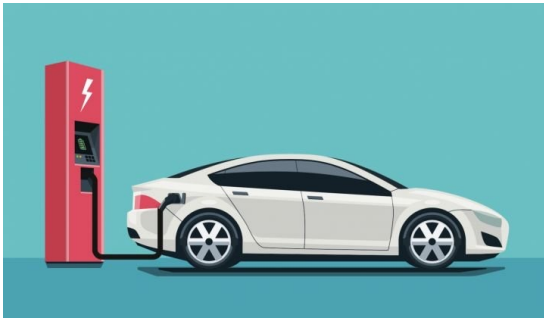
- The total energy of the universe is **constant**.
- Energy can be transferred from one form to another through heat and work.
- $\Delta E = q + w$
- Exothermic:  $q < 0$ ; Endothermic:  $q \geq 0$
- Work done **by** the system:  $w \leq 0$ ; Work done **on** the system:  $w \geq 0$

Work ( $w$ ): a force acting through a distance  $= F \times d$

**Energy**  $\longleftrightarrow$  **Energy**

Heat ( $q$ ): energy flow caused by a difference in temperature

*When an electric car discharge...*



# The First Law of Thermodynamics

*The law of conservation of energy:*

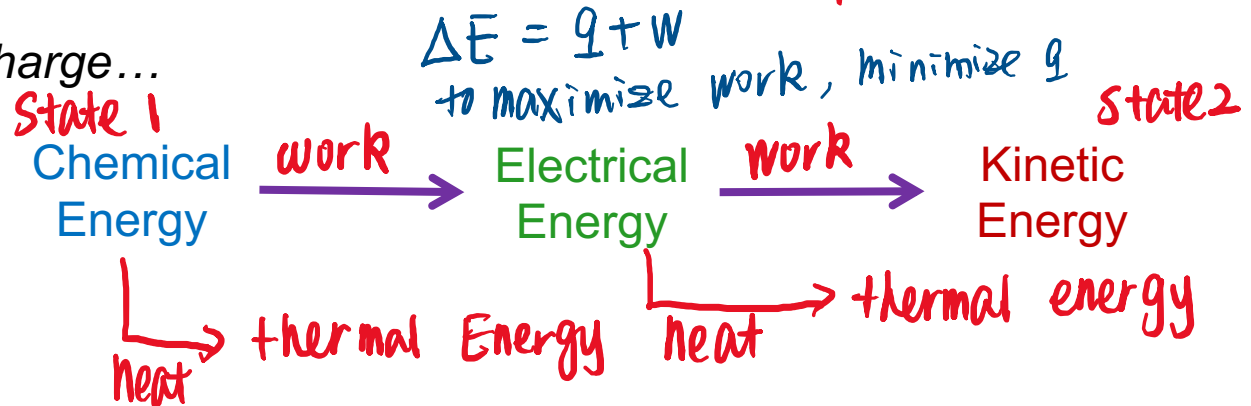
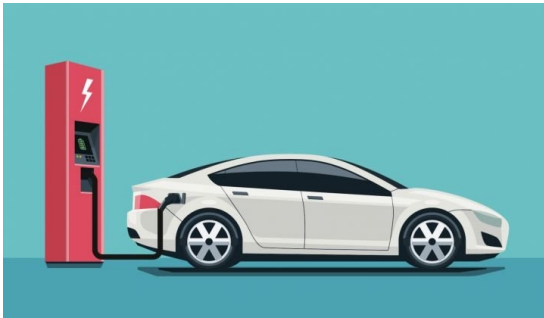
- The total energy of the universe is **constant**.
- Energy can be convert from one form to another through work and heat.
- $\Delta E = q + w$        $q = \Delta E - w$        $q$  is not a state function
- Exothermic:  $q \leq 0$ ; Endothermic:  $q \geq 0$
- Work done **by** the system:  $w \leq 0$ ; Work done **on** the system:  $w \geq 0$

Work ( $w$ ): a force acting through a distance



Heat ( $q$ ): energy flow caused by a difference in temperature

When an electric car discharge...



# Your Turn! $\Delta E = q + w$

Based on the first law of thermodynamics, which of the following four scenarios is **definitely** an **endothermic** process?

~~A~~ The internal energy increases for a given system.

$$\Delta E = q + w$$

15  
(+)

**B.** The internal energy increases within a given system while the system is doing work on its surroundings.

$$\Delta E = q + w > 0$$

100%  
(+) (+) (-)

C. The internal energy increases within a given system while work is being done on it.

$$\Delta E = q + w > 0$$

(+) (+) (-)

~~D~~ The internal energy decreases within a given system while work is being done on it.

$$\Delta E = q + w$$

(-) (-) (+)  
100%

~~E~~ None of the choices applies.



Exothermic:  $q < 0$ ; Endothermic:  $q > 0$

Work done by system:  $w < 0$ ; Work done on system:  $w > 0$

# Announcement 2/6

Assignments that are due soon

- Pre-Class Assignment  
Week 2-M

Office Hour

- Dr. Wang's:  
Fri 2-3 PM at KC 382



**Knight-Hennessy Scholars  
Stanford University**

**Information session**

**Date:** Friday, February 6, 2026  
**Time:** 12:00 PM  
**Location:** Beckman Hall 401

**Enjoy Pizza while learning  
about the program!**

Join a conversation with an admission officer from Knight-Hennessy Scholars, a multidisciplinary, multicultural graduate scholarship program.

Each Knight-Hennessy scholar receives up to three years of financial support to pursue graduate studies at Stanford. Scholars form a community while engaging in experiences that prepare them to be visionary, courageous, and collaborative leaders who address the world's complex challenges.

**Register:**



**kh.stanford.edu  
@KnightHennessy**

# Quantifying Heat (q)

## Heat Capacity (C)

- Different object has different ability to absorb thermal energy (heat)
- $C = \frac{q}{\Delta T}$
- $\Delta T = T_{\text{final}} - T_{\text{initial}}$
- Unit:  $\frac{\text{J}}{^\circ\text{C}}$ ;  $\frac{\text{J}}{\text{K}}$ ;  $\frac{\text{kJ}}{^\circ\text{C}}$
- Depends on identity and mass (extensive property) of the object.

## Specific Heat Capacity ( $C_s$ )

- $C_s = \frac{q}{m \times \Delta T}$
- Unit:  $\frac{\text{J}}{\text{g} \cdot ^\circ\text{C}}$ ;  $\frac{\text{J}}{\text{mol} \cdot \Delta T}$ ;  $\frac{\text{kJ}}{\text{g} \cdot ^\circ\text{C}}$
- Intensive property: Independent to mass

**TABLE 9.2 Specific Heat Capacities of Some Common Substances**

| Substance        | Specific Heat Capacity, $C_s$ (J/g · °C)* |
|------------------|-------------------------------------------|
| <b>Elements</b>  |                                           |
| Lead             | 0.128                                     |
| Gold             | 0.128                                     |
| Silver           | 0.235                                     |
| Copper           | 0.385                                     |
| Iron             | 0.449                                     |
| Aluminum         | 0.903                                     |
| <b>Compounds</b> |                                           |
| Ethanol          | 2.42                                      |
| Water            | 4.18 ←                                    |
| <b>Materials</b> |                                           |
| Glass (Pyrex)    | 0.75                                      |
| Granite          | 0.79                                      |
| Sand             | 0.84 ←                                    |

\*At 298 K.

# Quantifying Heat (q)

## Heat Capacity (C)

- Different object has different ability to absorb thermal energy (heat)
- $C = \frac{q}{\Delta T}$
- $\Delta T = T_{\text{final}} - T_{\text{initial}}$
- Unit: J/°C; kJ/°C; kJ/K
- Depends on identity and mass (extensive property) of the object.

## Specific Heat Capacity (C<sub>s</sub>)

- $C_s = \frac{q}{m \times \Delta T}$
- Unit: J/(g·°C); kJ/(g·°C); J/(g·K);
- Intensive property: Independent to mass

**TABLE 9.2 Specific Heat Capacities of Some Common Substances**

| Substance        | Specific Heat Capacity, C <sub>s</sub> (J/g · °C)* |
|------------------|----------------------------------------------------|
| <b>Elements</b>  |                                                    |
| Lead             | 0.128                                              |
| Gold             | 0.128                                              |
| Silver           | 0.235                                              |
| Copper           | 0.385                                              |
| Iron             | 0.449                                              |
| Aluminum         | 0.903                                              |
| <b>Compounds</b> |                                                    |
| Ethanol          | 2.42                                               |
| Water            | 4.18                                               |
| <b>Materials</b> |                                                    |
| Glass (Pyrex)    | 0.75                                               |
| Granite          | 0.79                                               |
| Sand             | 0.84                                               |

\*At 298 K.

# Your Turn! $C_s = \frac{q}{m \times \Delta T}$

Which sample is most likely to undergo the smallest change in temperature upon the absorption of 100 kJ of heat?

- A. 15 g water
- B. 15 g lead
- ☒ C. 50 g water
- D. 50 g lead



$$\downarrow \Delta T = \frac{q}{\uparrow m \times C_s \uparrow}$$

**TABLE 9.2 Specific Heat Capacities of Some Common Substances**

| Substance        | Specific Heat Capacity, $C_s$ (J/g · °C) * |
|------------------|--------------------------------------------|
| <b>Elements</b>  |                                            |
| Lead             | 0.128                                      |
| Gold             | 0.128                                      |
| Silver           | 0.235                                      |
| Copper           | 0.385                                      |
| Iron             | 0.449                                      |
| Aluminum         | 0.903                                      |
| <b>Compounds</b> |                                            |
| Ethanol          | 2.42                                       |
| Water            | 4.18                                       |
| <b>Materials</b> |                                            |
| Glass (Pyrex)    | 0.75                                       |
| Granite          | 0.79                                       |
| Sand             | 0.84                                       |

\*At 298 K.



# Your Turn! $q = C \times \Delta T = C_s \times m \times \Delta T$

- a) A cup of water is studied as a system and its heat capacity is known to be  $71.2 \text{ J/}^\circ\text{C}$ . How much heat will the cup of water release if its temperature changed from  $23.5^\circ\text{C}$  to  $19.2^\circ\text{C}$ ?
- b) The heat released in (a) was completely absorbed by a block of iron metal. The mass of the iron metal is  $5.12 \text{ g}$  and the specific heat capacity of iron is  $0.449 \text{ J/(g}\cdot^\circ\text{C)}$ . Before absorbing the heat, the temperature of iron is  $25^\circ\text{C}$ . What is the final temperature of iron metal?

# Your Turn! $q = C \times \Delta T = C_s \times m \times \Delta T$

- a) A cup of water is studied as a system and its heat capacity is known to be  $71.2 \text{ J/}^\circ\text{C}$ . How much heat will the cup of water release if its temperature changed from  $23.5^\circ\text{C}$  to  $19.2^\circ\text{C}$ ? *exothermic*

$$q_w = \frac{71.2 \text{ J}}{1^\circ\text{C}} \times (19.2^\circ\text{C} - 23.5^\circ\text{C}) = -306 \text{ J}$$

↓

- b) The heat released in (a) was completely absorbed by a block of iron metal. The mass of the iron metal is  $5.12 \text{ g}$  and the specific heat capacity of iron is  $0.449 \text{ J/(g}\cdot^\circ\text{C)}$ . Before absorbing the heat, the temperature of iron is  $25^\circ\text{C}$ . What is the final temperature of iron metal?

$$q_{\text{iron}} = -q_{\text{water}} = +306 \text{ J} = \frac{0.449 \text{ J}}{\text{g}\cdot^\circ\text{C}} \times 5.12 \text{ g} \times (T_F - 25^\circ\text{C})$$

$$T_F - 25^\circ\text{C} = 133^\circ\text{C}$$

$$T_F = 158^\circ\text{C}$$

# Your Turn! $q = C \times \Delta T = C_s \times m \times \Delta T$

- c) A 32.5 g cube of Al initially at 45.8 °C is submerged into 105.3 g of water at 15.4 °C. The specific heat capacity of Al and water are 0.903 J/(g·°C) and 4.18 J/(g·°C). What is the final temperature of both substances at thermal equilibrium? Assume the heat transfer only happens Al and water.

|                |                                                                                   |                   |                                                                                               |
|----------------|-----------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------|
|                | <div style="border: 1px solid red; padding: 2px; display: inline-block;">Al</div> | $\xrightarrow{q}$ | <div style="border: 1px solid red; padding: 2px; display: inline-block;">H<sub>2</sub>O</div> |
| m              | 32.5g                                                                             |                   | 105.3g                                                                                        |
| T <sub>i</sub> | 45.8°C                                                                            |                   | 15.4°C                                                                                        |
| C <sub>s</sub> | 0.903 J/g·°C                                                                      |                   | 4.18 J/g·°C                                                                                   |
| T <sub>F</sub> | T <sub>F-Al</sub> = T <sub>F-H<sub>2</sub>O</sub>                                 |                   |                                                                                               |

$$q_{Al} + q_{H_2O} = 0$$
$$q_{Al} = -q_{H_2O}$$

---

# Your Turn! $q = C \times \Delta T = C_s \times m \times \Delta T$

- c) A 32.5 g cube of Al initially at 45.8 °C is submerged into 105.3 g of water at 15.4 °C. What is the final temperature of both substances at thermal equilibrium? Assume the heat transfer only happens Al and water.

$\boxed{\text{Al}} \xrightarrow{q} \boxed{\text{H}_2\text{O}}$  at the thermal equilibrium:

|                                             |                                       |                                                                                                                                                                                      |
|---------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $m: 32.5 \text{ g}$                         | $105.3 \text{ g}$                     | $\begin{cases} T_{\text{Al Final}} = T_{\text{H}_2\text{O Final}} = T_f \\ q_{\text{Al}} = - q_{\text{H}_2\text{O}} \text{ or } -q_{\text{Al}} = q_{\text{H}_2\text{O}} \end{cases}$ |
| $T_i: 45.8^\circ\text{C}$                   | $15.4^\circ\text{C}$                  |                                                                                                                                                                                      |
| $C_s: 0.903 \text{ J/g}\cdot^\circ\text{C}$ | $4.18 \text{ J/g}\cdot^\circ\text{C}$ |                                                                                                                                                                                      |

$$\underbrace{\frac{0.903 \text{ J}}{\text{g}\cdot^\circ\text{C}} \times 32.5 \text{ g} \times (T_f - 45.8^\circ\text{C})}_{q_{\text{Al}}} = - \underbrace{\frac{4.18 \text{ J}}{\text{g}\cdot^\circ\text{C}} \times 105.3 \text{ g} \times (T_f - 15.4^\circ\text{C})}_{-q_{\text{H}_2\text{O}}}$$

$$\frac{29.35 \text{ J}}{^\circ\text{C}} \times (T_f - 45.8^\circ\text{C}) = - \frac{440.2 \text{ J}}{^\circ\text{C}} \times (T_f - 15.4^\circ\text{C})$$

$$\frac{469.6 \text{ J}}{^\circ\text{C}} T_f = 6779 \text{ J} + 1344 \text{ J}$$

$$T_f = 17.3^\circ\text{C}$$

# Quantifying work (w)

- In general,  $w = F \times d$

## Pressure-Volume Work (P-V work)

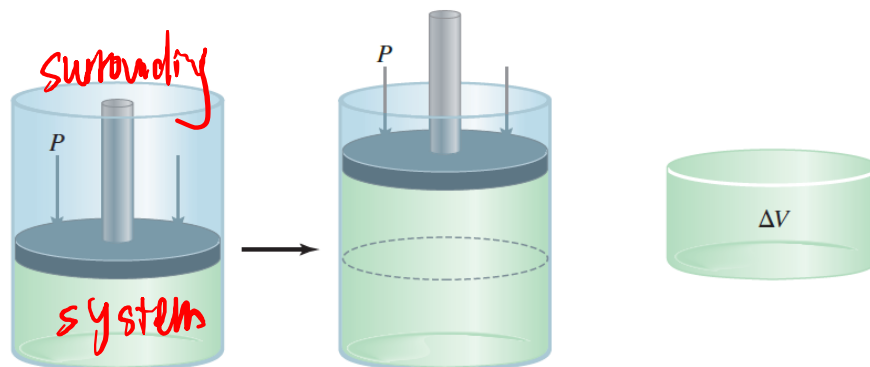
- In Chem 150, work only refers to compression or expansion of gas against constant external pressure:

surroundings

$$w = -P_{\text{external}} \times \Delta V$$

$$= - \frac{F_{\text{ext}}}{A} \times \Delta(A \times h)$$

$$101.3 \text{ J} = 1 \text{ L} \cdot \text{atm}$$



Consider gas as the system:

- Gas **contracts**:

The volume of gas  $\downarrow$  ( $\Delta V < 0$ ),  $w = -P \times \Delta V > 0$ : work is being done on the system

- Gas **expands**:

The volume of gas  $\uparrow$  ( $\Delta V > 0$ ),  $w = -P \times \Delta V < 0$ : work is being done by the system

# Quantifying work (w)

- $w = F \times d$

## Pressure-Volume Work (P-V work)

- Pressure ( $P$ ) of a gas is defined as **force** ( $F$ ) applied per unit **area** ( $A$ ):

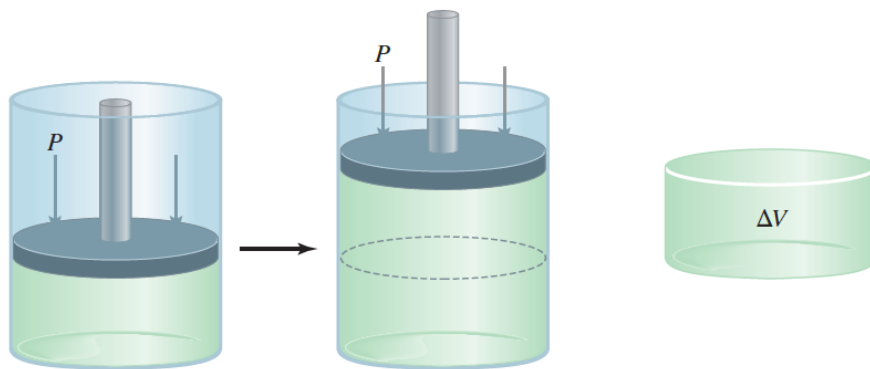
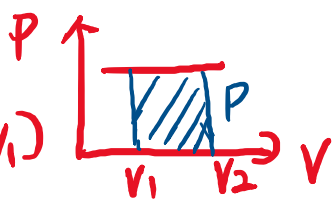
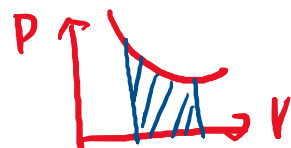
$$P = \frac{F}{A} \quad \underline{F = P \times A} \rightarrow w =$$

- In Chem 150, work only refers to compression or expansion of gas under constant pressure:

$$w = P \times A \times d = P \times V$$

$$w = -P \times \Delta V = -P \times (V_2 - V_1)$$

$$101.3 \text{ J} = 1 \text{ L} \cdot \text{atm}$$



Consider gas as the system:

- Gas **contracts**:

The volume of gas  $\downarrow$  ( $\Delta V \leq 0$ ),  $w = -p \times \Delta V \geq 0$ : work is being done **on** the system

- Gas **expands**:

The volume of gas  $\uparrow$  ( $\Delta V \geq 0$ ),  $w = -p \times \Delta V \leq 0$ : work is being done **by** the system

# Your Turn! $w = -P \times \Delta V$ ; $\Delta E = q + w$

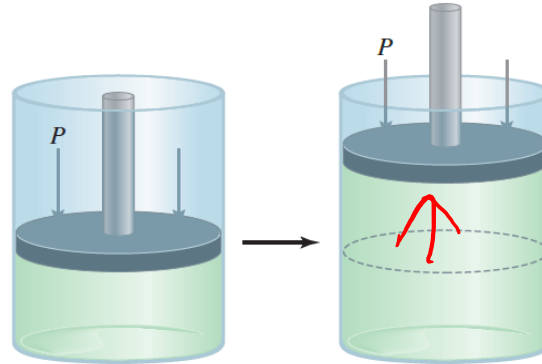
Gases push up a cylinder in a car engine. (Gases are the system.) If no heat is transferred during the process, the sign of work (relative to the system) is \_\_\_\_\_ and the system \_\_\_\_\_ energy.

A. negative; release

B. positive; release

C. negative; absorb

D. positive; absorb



# Announcements 2/9

Assignments that are due soon

- Pre-Class Assignment Week 2-W

Office Hour

- Dr. Wang's: Tue 10-11 AM at KC 230
- Paige Kwan's: Mon 11:30 AM -12:30 PM at the TLC (Demille 160P)
- Saige Douglass's: Tue ~~5:30 - 6:30~~ PM at the TLC (Demille 160P)

12:00 - 1:00



# Measuring $\Delta E$ for Chemical Reactions

Coffee-cup (styrofoam) calorimeters:

- The calorimeter is open to the atmosphere and the **pressure is constant**.

- System: reaction

- Surroundings: calorimeter ← surr.

- Knowing:  $T_{\text{initial}}$ ,  $T_{\text{final}}$ ,  $C_{\text{cal}}$ ,  $P_{\text{atm}}$

- Goal: Calculate  $\Delta E_{\text{rxn}} = q_{\text{rxn}} + w$   
 $= q_p - P_{\text{atm}} \times \Delta V$

No heat escaped:  $q_{\text{cal}} + q_p = 0$   $q_p = -q_{\text{cal}}$

Heat change for calorimeter:  $q_{\text{cal}} = C_{\text{cal}} \times (T_F - T_I)$

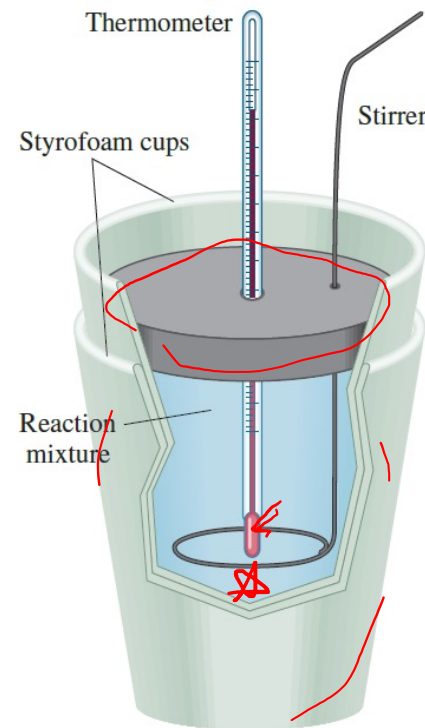
**Exothermic** reaction  $\leftrightarrow$  temperature of the calorimeter  $\uparrow$

**Endothermic** reaction  $\leftrightarrow$  temperature of the calorimeter  $\downarrow$

- Therefore,  $\Delta E_{\text{rxn}} =$   $\rightarrow$  heat capacity for calorimeter  
 $= - (C_{\text{cal}} \times (T_F - T_I)) - P_{\text{atm}} \times \Delta V$   $\leftarrow \Delta n(g_{\text{wb}})$

- For reactions involve only liquid and gas:

No gas evolves and the P-V work is negligible:  $\Delta E_{\text{rxn}} \approx - C_{\text{cal}} \times (T_F - T_I)$



# Measuring $\Delta E$ for Chemical Reactions

Coffe-cup (styrofoam) calorimeters:  $q_p$

- The calorimeter is open to the atmosphere and the pressure is constant.
- System: rxn
- Surroundings: calorimeter
- Knowing:  $T_{\text{initial}}$ ,  $T_{\text{final}}$ ,  $C_{\text{cal}}$ ,  $P$
- Goal: Calculate  $\Delta E_{\text{rxn}} = q_{\text{rxn}} + w$

$$= q_p - P\Delta V$$

- Heat absorbed by calorimeter:

$$q_{\text{cal}} = C_{\text{cal}} (T_F - T_I)$$

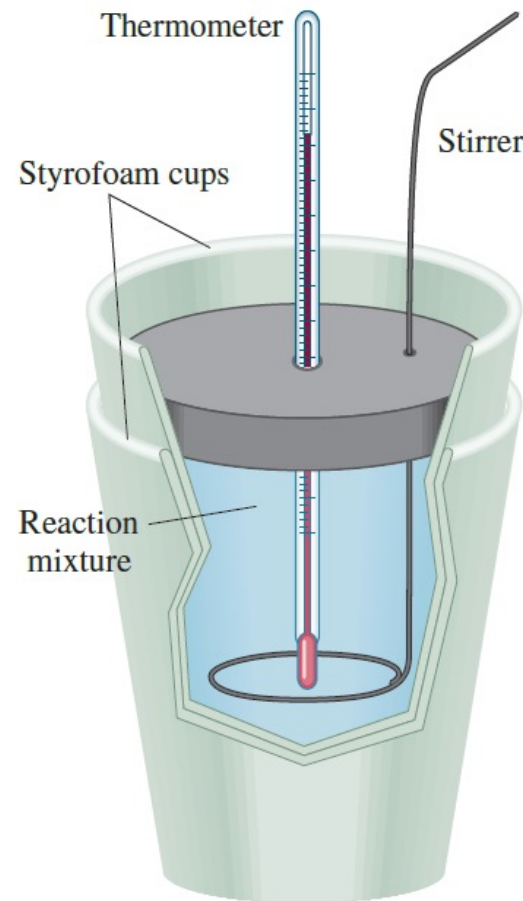
No heat escaped:  $q_{\text{cal}} + q_p = 0$

$$q_p = -q_{\text{cal}} = -C_{\text{cal}} (T_F - T_I)$$

- Therefore,  $\Delta E_{\text{rxn}} = -C_{\text{cal}} (T_F - T_I) - P\Delta V$

- For reactions involving only liquid and gas:  $\Delta V = 0$   $w = -P\Delta V = 0$

No gas evolves and the P-V work is negligible:  $\Delta E_{\text{rxn}} \approx q_p$



# Your Turn! $q = C \times \Delta T = C_s \times m \times \Delta T$

Dissolving  $\text{NH}_4\text{NO}_3$  in water is a process found in cold packs. When 20.0 g of  $\text{NH}_4\text{NO}_3$  was rapidly dissolved in 75.0 mL of  $\text{H}_2\text{O}$  in a coffee cup calorimeter, the temperature dropped from 25.7 °C to 10.4 °C. The specific heat of the solution is 4.18 J/g°C and the density of  $\text{H}_2\text{O}$  is 0.993 g/mL. What is the value of  $\Delta E$  for the reaction?

a) Is the system under CONSTANT PRESSURE or CONSTANT VOLUME?

b) Is work done? YES or NO  $w = -P\Delta V$



c) Is the reaction ENDOTHERMIC or EXOTHERMIC?

|                        |             | $\Delta T$ ( $T_{\text{final}} - T_{\text{initial}}$ ) | q (heat) | Endo-/exo- |
|------------------------|-------------|--------------------------------------------------------|----------|------------|
| system<br>$\uparrow q$ | reaction    |                                                        | (+)      | endo -     |
| surrounding            | calorimeter | $\downarrow$                                           | (-)      | exo -      |

d) Does heat flow from system to surroundings OR surroundings to system?

## Your Turn! $q = C \times \Delta T = C_s \times m \times \Delta T$

Dissolving  $\text{NH}_4\text{NO}_3$  in water is a process found in cold packs. When 20.00 g of  $\text{NH}_4\text{NO}_3$  was rapidly dissolved in 75.00 mL of  $\text{H}_2\text{O}$  in a coffee cup calorimeter, the temperature dropped from 25.70 °C to 10.40 °C. The specific heat of the solution is 4.18 J/g°C and the density of  $\text{H}_2\text{O}$  is 0.9930 g/mL. What is the value of  $\Delta E$  for the reaction?

$$\Delta E = q_{\text{rxn}} - P_{\text{ext}} \Delta V$$

$$\stackrel{\Delta V \approx 0}{\Rightarrow} q_{\text{rxn}} = -q_{\text{cal}}$$

$$= -4.18 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}} \times (20.00 \text{ g} + 75.00 \text{ mL} \times \frac{0.9930 \text{ g}}{\text{mL}}) \times (10.40^\circ\text{C} - 25.70^\circ\text{C})$$

$$= +6040 \text{ J}$$

## Your Turn! $q = C \times \Delta T = C_s \times m \times \Delta T$

Dissolving  $\text{NH}_4\text{NO}_3$  in water is a process found in cold packs. When 20.00 g of  $\text{NH}_4\text{NO}_3$  was rapidly dissolved in 75.00 mL of  $\text{H}_2\text{O}$  in a coffee cup calorimeter, the temperature dropped from 25.70 °C to 10.40 °C. The specific heat of the solution is 4.18 J/g°C and the density of  $\text{H}_2\text{O}$  is 0.9930 g/mL. What is the value of  $\Delta E$  for the reaction?

$$\Delta E_{\text{rxn}} = q_{\text{rxn}} + w \stackrel{w=0}{=} q_{\text{rxn}} = -q_{\text{cal}}$$

$$= - \frac{4.180 \text{ J}}{\cancel{\text{g} \cdot ^\circ\text{C}}} \times \left( 20.00 \cancel{\text{g}} + 75.00 \cancel{\text{mL}} \times \frac{0.9930 \cancel{\text{g}}}{\cancel{\text{mL}}} \right) \times (10.40 \cancel{^\circ\text{C}} - 25.70 \cancel{^\circ\text{C}})$$

$$= +6042 \text{ J}$$

The reaction is endothermic and (+) sign make sense.